Ben Altangerel

+44 7984 611311 | bayasal@hotmail.co.uk | linkedin.com/in/bayasalaltangerel | baltangerel.github.io | London

Technical Skills

Simulation: ANSYS Fluent (CFD), ANSYS Mechanical (FEA), ParaView, mesh convergence and validation
Design: SolidWorks, Fusion 360, BS 8888 drawings, GD&T, design for manufacture
Programming: Python, MATLAB, C#, Unity, Git
Hands-on: Laser cutting, 3D printing, soldering, ArduPilot, flight testing

Projects

Understanding and Simulating Flows Through VR Sep. 2024 – Jun. 2025
MEng Dissertation Project (First Class – 71%) *Sheffield*

- Built an interactive VR environment in Unity and C# with grabbable contour planes, animated particle flow and a velocity probe, by creating a Fluent → ParaView → Blender → Unity pipeline with a custom shader matching the CFD colour data.
- Produced CFD results accurate enough to teach from, with lift within 1.2% of wind tunnel data at Mach 0.84, by setting up and validating ONERA M6 wing simulations in ANSYS Fluent against experimental pressure data at 7 spanwise sections.
- Animated around 70,000 streamline particles in VR, with play, pause and restart controls, by writing C# scripts that read particle data exported from Fluent and reuse objects through a pooled particle system.

Aerospace Design III – Transitioning Fixed-Wing UAS Sep. 2023 – Jun. 2024
Project Manager *Sheffield*

- Delivered a 2.1 kg, 1.4 m-span VTOL quadplane £153 (19%) under its £800 budget and two weeks ahead of the testing milestone, by leading a six-person team with a week-by-week project plan, risk register and fortnightly client reports.
- Secured the team's Flight Test Certificate by owning the safety case and all formal documents, then diagnosed a flight-day crash from onboard video and flight logs and had the aircraft repaired and flying again the same day.
- Brought the project back on schedule after a weak start by introducing purposeful weekly meetings and rebuilding the plan with an owner, priority and deadline for every task, reaching the manufacture gate one week early.

Engine Mount Ageing – FEA Validated Against Experiment Sep. 2024 – Dec. 2024
Individual Project *Sheffield*

- Showed that 130,000 km of service raises peak strain in a rubber engine mount by 50% (0.30 to 0.45), giving evidence for setting replacement intervals, by building static structural ANSYS models of a new and an aged mount.
- Matched measured strain to within 9% (0.038 predicted against about 0.035 measured), by validating both models against digital image correlation data from published lab tests.
- Kept results mesh-independent (1.2% change between 3 mm and 2 mm meshes) while modelling only a quarter of the mount, by using symmetry boundary conditions and refining only the high-strain regions with spheres of influence.

Lightweight Plywood Crane Feb. 2023 – May 2023
Structures Lead *Sheffield*

- Placed 3rd of 42 teams for the lightest structure to lift an 18 kg mass, by designing both of the team's laser-cut plywood cranes, which lifted the load with no fractures or buckling.
- Raised a design's buckling load from 0.28× to 4.5× the applied load (safety factor 3.4 at 129 g) by adding interlocking cross-supports, after Fusion 360 FEA showed buckling, not stress, was the limiting failure mode.
- Cut the lightweight design to 54 g at a 1.8 safety factor by using stress plots to relocate under-loaded material, with plywood properties from our own tensile tests and mesh convergence within 3%.

Education

The University of Sheffield Sheffield
Master of Engineering in Aerospace Engineering *Upper Second-Class Honours (2:1)*

- Relevant modules: Design of Structures, Machines and Systems (75%), Advanced Aerospace Propulsion Technology (74%), Aerostructures (73%), Control Systems Design and Analysis (69%)

Loreto College Manchester
A-Levels: Law (A), Physics (B), Mathematics (B) *Sep. 2019 – Jun. 2021*

Work Experience

Waiter and Front of House Nov. 2024 – Present
BOX (ARC Inspirations) *Manchester & Sheffield*

- Received the most positive customer reviews on the team during the World Cup (93) by proactively handling queries and prioritising orders at peak service.
- Kept service accurate and on time during busy periods by communicating clearly with managers, kitchen staff and co-workers, relaying changes and flagging issues early.

Supervisor Jul. 2022 – Oct. 2024
Fuku *Manchester*

- Cut the restaurant's supply costs by 12% by approaching potential suppliers, comparing their prices and negotiating better rates.
- Kept the restaurant fully stocked for every service by carrying out regular stock counts and keeping inventory records accurate and up to date.

Bartender and Back of House Jun. 2021 – Jul. 2022
Turtle Bay *Manchester & Sheffield*

- Recommended, prepared and served drinks and cocktails to the guests, as requested, with pace and accuracy.
- Maintained the cleanliness of the bar and equipment to ensure all drinks are served to the highest quality.

Career Break – COVID-19 Restrictions Apr. 2020 – May 2021
Full-time A-Level Student *Manchester*

- Out of work during UK COVID-19 restrictions, from the first national lockdown on 23 March 2020 until indoor hospitality reopened on 17 May 2021, while completing A-levels at Loreto College.

Retail Assistant Apr. 2019 – Mar. 2020
Topshop and Topman *Manchester*

- Operated tills and assisted customers on the shop floor, delivering friendly and attentive customer service.
- Showcased attention to detail and a proactive work ethic by maintaining store cleanliness and organising inventory.

Laboratory Assistant May 2019
University of Manchester Biology Research *Manchester*

- Logged 60 experimental results into the lab's spreadsheet on the day they were produced, checking each against the paper record and correcting 3 transcription errors.
- Counted and updated all 120 items on the lab inventory by midweek and flagged 5 below minimum stock with the quantities needed, so the order was placed before the next experiment.
- Acted as first point of contact for around 10 calls a day and managed 6 lab bookings and 4 staff meetings, resolving 1 double booking before it affected an experiment.

Extra-Curricular Activities

Football

- Organiser and treasurer for a workplace football team, running weekly matches and finances. 2025 – Present
- Won a year-long 5-a-side football knockout competition organised by the University of Sheffield. 2023 – 2024

Volunteer Apr. 2026 – Present
Hackspace Manchester *Manchester*

- Volunteer at a community makerspace, running training sessions and tours across laser cutting, 3D printing, electronics and metalwork workshops.